Comparable evaluation of occluded 6D object poses in RGB D benchmarks

Literature Review

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Abstract

Background. Occlusion is a central difficulty in six-degree-of-freedom (6D) object pose estimation, but an improvement on a cluttered color-and-depth (RGB-D) dataset does not by itself establish greater occlusion robustness. Visibility annotations, object symmetries, available inputs and score aggregation determine what a reported result means.

Survey methodology. This critical methodological review examines primary benchmark papers, public evaluation code, dataset documentation and representative pose estimators, supported by a documented targeted search completed on 19 October 2026. It addresses rigid instance-level poses in settings associated with the Benchmark for 6D Object Pose Estimation (BOP). Thirty scholarly sources and four documentation records support the synthesis; the search is selective rather than exhaustive, and no new model evaluation or pooled performance estimate is reported.

Synthesis. We organize comparability into four layers: observed visibility, admissible pose equivalence, task information and score aggregation. The framework distinguishes projected visibility from missing depth and truncation, global symmetry from observation-dependent ambiguity, and localization recall from detection precision. A worked documentary audit shows how segmentation assistance, refinement, object onboarding and changes in ground-truth equivalence can invalidate apparent cross-paper comparisons. Recent distribution-aware and visibility-balanced evaluations address different parts of this problem and should be interpreted separately.

Conclusions. Claims about occlusion robustness should identify the visibility measure and evaluated population, preserve the task and equivalence rules, and report conditional performance alongside the official benchmark score. We provide an auditable comparison record, a decision procedure and a minimum reporting checklist. These support more defensible literature synthesis and benchmark design without treating pose accuracy as evidence of manipulation success.

Introduction

Estimating the rigid transformation from an object model to the camera is a recurring component of robotic perception. A manipulator often observes only a fragment of the object it must grasp, inspect or assemble. The practical question is therefore not simply which estimator has the highest published score, but which evidence supports reliable inference from partial observations. The distinction matters when comparing results from different datasets, modalities and benchmark generations. Detecting more easy instances, adding depth during refinement or supplying stronger object information can improve an aggregate score. These changes need not improve performance on severely occluded targets.

The Benchmark for 6D Object Pose Estimation (BOP) has already made major progress toward common evaluation. Its original benchmark unified datasets and ambiguity-aware scoring, and subsequent challenges established shared protocols and public evaluation tools (Hodaň et al., 2018; Hodaň et al., 2020; Sundermeyer et al., 2023). These contributions are the starting point of this review. The remaining difficulty is interpreting results whose protocols differ or whose

common benchmark score answers a broader question than occlusion robustness. Standardizing a pose error does not, by itself, standardize the distribution of visible evidence or the information supplied to the estimator.

Existing reviews provide broad accounts of object pose recovery, instance- and category-level learning, and application-driven model selection (Sahin et al., 2020; Guan et al., 2024; Ordoumpozanis & Papakostas, 2025). The recent RGB survey by Aguirrezabal, Aguinaga and Alvarez-Gila emphasizes augmented reality, runtime and modern benchmarks (Aguirrezabal, Aguinaga & Alvarez-Gila, 2026). These works also discuss occlusion, symmetry and metrics. We therefore do not claim that those issues have been overlooked. Our narrower organizing question is: under which documented conditions can two reported results support a comparison of occlusion robustness? Table 1 positions this question against the closest retrieved reviews.

[Table 1 about here]

Two recent contributions make this question particularly timely. BOP-Distrib introduces image-wise ambiguity annotations and evaluates distributions of plausible poses, while WALDO proposes visibility-decile aggregation to reduce dependence on the observed visibility distribution (Meden et al., 2026; Pakdamansavoji et al., 2026). Their evaluation contributions need to be considered alongside their estimator results. They intervene at different points in evaluation: one changes the acceptable explanations of an observation; the other changes how performance over visibility levels is summarized.

This review makes three contributions. First, it separates four layers that must be checked before interpreting a cross-paper difference: visibility, pose equivalence, task information and aggregation. Second, it applies those layers in a worked documentary audit of representative published protocols, identifying what comparisons are justified and what additional artifacts would be needed. Third, it provides a reusable reporting record and decision procedure for researchers and reviewers. These are methodological synthesis tools, not new pose metrics, a new dataset or an experimentally validated standard.

Survey methodology

Scope and retrieval

We conducted a critical methodological review with a documented targeted search, completed on 19 October 2026. The scope is rigid, instance-level object pose estimation in RGB-D benchmark settings associated with BOP, including RGB-only and multi-view variants when they reveal a material difference in evaluation information. Category-level shape and scale estimation, articulated objects, human pose estimation, long-horizon tracking and end-to-end grasp learning are outside the principal synthesis. Novel-object methods are included when the provided model or reference observations define an instance-level target.

Discovery used web searches directed toward publisher records, the Computer Vision Foundation proceedings, arXiv and official BOP pages. Search families combined “6D object pose estimation” with “survey”, “occlusion”, “RGB-D”, “symmetry”, “visibility”, “BOP”,

"evaluation" or "metrics". Exact-title searches resolved benchmark reports, datasets and methods. Reference following from the BOP reports and retrieved reviews supplied additional methodological sources. Supplementary File S2 records the actual queries, source routes and access failures; the archived research record retains the returned search outputs. We did not search subscription bibliographic databases, register a prospective protocol or use citation counts as an inclusion threshold.

Sources were retained if they defined a relevant dataset or evaluation rule, analyzed ambiguity or visibility, or documented an estimator setting needed to expose a comparability issue. Representative estimators were chosen to cover visible-point voting, RGB-D fusion, regression, rendering-based refinement, foundation-feature matching and pose distributions. This purposive choice is not a census of the field. The final scholarly corpus contains 30 works, including four broad reviews used for positioning, together with four consolidated BOP documentation records. This is a count of sources used in the synthesis, not a database screening total. No PRISMA flow, exclusion count or inter-reviewer agreement is asserted.

Evidence extraction and boundaries

The extraction record identifies the source and version, relevant section, supported claim, modality, task, assistance, error function and limitation. Primary papers support descriptions of their own methods; official benchmark documentation and public code support implementation details. Bibliographic identities were checked against primary records and the available full texts. Publisher proceedings take priority over preprint metadata when both are available; one challenge report is cited through its verified arXiv record. Access limitations and corrected metadata discrepancies are retained in the research notes. Supplementary File S1 links each source to the conclusions it supports.

We read the relevant benchmark definitions, dataset descriptions, method and evaluation sections, and the associated tables or figure captions used in the audit. Public BOP toolkit files were inspected at a recorded repository revision, with file hashes retained. Documentation is evidence of how an evaluation is specified or implemented, not independent evidence of its empirical superiority. Extraction was not independently duplicated. We therefore avoid prevalence estimates about reporting quality, causal claims about architectural improvements and assertions of exhaustive coverage. No benchmark image set was downloaded for evaluation, no estimator was executed and no published score was converted into a common ranking.

For synthesis, we distinguished source-defined facts, deductions from explicit definitions, and recommendations proposed here. Direct comparability requires agreement on the quantities that define the comparison, not identical neural architectures. When a necessary field is unreported, it is marked unresolved rather than inferred from the method name. The worked audit is documentary: it tests whether a claim can be justified from the reported protocol, not whether an estimator performs as its authors report.

Four layers of comparability

A result is an output of a measurement procedure. We represent its evaluation conditions by C = (V, E, I, A). Here V describes visibility, E admissible pose equivalence, I allowed information and resources, and A aggregation and matching. This record complements BOP metadata and evaluation rules; it does not replace them. Its purpose is to keep a literature comparison attached to the conditions under which its scores were obtained (Figure 1).

[Figure 1 about here]

Visibility is an annotation dependent observation

Let T map model points into the camera frame. Let M_{full} be the isolated-object silhouette at that pose, extended to include the truncated part outside the image. Let M_{in} be the silhouette inside the image and M_{vis} the pixels considered visible by a specified visibility test. For nonempty denominators, define

$$v_{\mathrm{BOP}} = \frac{|M_{\mathrm{vis}}|}{|M_{\mathrm{full}}|}, \quad v_{\mathrm{in}} = \frac{|M_{\mathrm{vis}}|}{|M_{\mathrm{in}}|}, \quad t = \frac{|M_{\mathrm{in}}|}{|M_{\mathrm{full}}|} \quad (1)$$

$$v_{\mathrm{BOP}} = t\, v_{\mathrm{in}}$$

The first ratio follows the BOP annotation fields px_count_visib/px_count_all; the inspected calculation script explicitly counts the full silhouette including truncation (BOP contributors, 2026d). The factorization is a deduction from these set definitions. It explains why a low stored visibility fraction need not mean that other objects cover the same proportion of the in-frame silhouette. A study of inter-object occlusion should either control truncation or report it separately. Equation 1 should not be retroactively imposed on a dataset whose own annotation convention uses a different denominator.

These quantities concern projected pixels. They are not fractions of total three-dimensional surface area. Self-occlusion already determines which surface forms the isolated-object rendering at a viewpoint. External occlusion removes part of that view, while truncation removes part of the projection from the sensor field of view. Occlusion by a hand, a near-identical object or an unrelated background object may have equal projected area but leave very different identity and orientation cues. Reporting one visibility scalar cannot preserve those distinctions.

Depth validity is another variable. A pixel with no measured depth is not necessarily hidden by another object. The public toolkit distinguishes older and later visibility modes (BOP contributors, 2026d). The older mode treats missing-depth pixels as invisible; the later mode may consider them visible to accommodate poorly captured surfaces. Consequently, visible-mask area and valid-depth area are not interchangeable. A depth-completion module can change the estimator input without changing the official visibility annotation; recomputing visibility from the completed depth would change the measurement definition as well.

For occlusion analysis we recommend retaining the benchmark visibility field and its parameters, and supplementing it with truncation and in-frame depth validity when available. A visibility-stratified performance curve must specify its instance population, bin boundaries and counts.

Excluding highly occluded targets before evaluation limits the domain of the resulting claim, even if the remaining test images look cluttered. The BOP challenge definitions include visibility eligibility rules, so these rules should be read with the task specification rather than inferred from the dataset name (Nguyen et al., 2025).

Pose equivalence depends on what is being judged

For an object with declared global symmetries S, several rigid transformations can describe equivalent object configurations. The relevant target is then an equivalence class of poses, {Tg : g in S}, rather than a unique rigid transformation. Symmetry-aware evaluation of industrial objects predates recent foundation models (Brégier et al., 2017). The precise symmetry set still matters: geometric symmetry may be broken by texture, and approximate manufacturing detail may be retained or ignored depending on the benchmark convention.

Global equivalence differs from ambiguity induced by the current observation. An asymmetric object can appear rotationally ambiguous when its distinctive handle is hidden. Conversely, texture can disambiguate poses whose depth renderings coincide. Multi-hypothesis estimation makes this distinction explicit (Manhardt et al., 2019). BOP-Distrib addresses the evaluation side by defining image-wise pose distributions, demonstrating that changing the ambiguity annotation can change method rankings (Meden et al., 2026). The review treats that result as evidence about dependence on annotation, not as a newly reproduced ranking.

Three objectives should therefore be distinguished. Physical pose recovery asks which configuration actually occurred. Observation consistency asks which poses could explain the available sensor evidence. Task sufficiency asks which remaining pose uncertainty permits the intended action. These objectives can agree on an easy textured object and diverge on a partly visible symmetric part. A benchmark must choose an operational target; a literature review must preserve that choice. None of the three objectives is universally preferable.

Information conditions include the whole pipeline

In BOP localization, a method receives identities and numbers of target instances; detection removes that per-image object information. The 2023 challenge introduced constrained onboarding for unseen objects (Hodaň et al., 2024). Model-based novel-object tasks provide models during onboarding, while model-free tasks use reference observations under their own rules (BOP contributors, 2026c). A label such as “zero-shot” does not fully identify these conditions. It should be accompanied by the training exposure, object onboarding, model availability and detector provenance of the evaluated system.

The same discipline applies to sensor use. “RGB method” can describe an initialization stage even if a reported variant later uses depth for alignment. A single-view estimator can be embedded in a multi-view system. Ground-truth boxes, predicted masks and default benchmark detections define different sources of assistance. These are not minor implementation choices when an occlusion-related failure occurs before pose estimation. They determine whether the score describes an assisted pose stage or a full perception pipeline.

Aggregation defines the population represented by a score

A score may summarize instances, objects, scenes, datasets, error thresholds or visibility strata. These averaging operations answer different questions. A benchmark-wide average can be appropriate for its target distribution while providing little information about the least visible objects. Similarly, equal weighting of datasets prevents large datasets from dominating, but does not equalize visibility within each dataset. An occlusion claim must therefore name both the pose error and the population weighting.

This yields three possible comparison decisions. A direct comparison is defensible when the necessary conditions agree and all material fields are documented. A comparison requiring re-evaluation may be possible when saved predictions and annotations permit the missing stratification or metric to be recomputed. A contextual comparison is appropriate when the systems answer different questions, or when indispensable information is unavailable. A single incompatible field can be decisive; adding up the number of matching fields is not a valid comparability score.

Datasets as sources of different occlusion evidence

BOP-Classic-Core brings together LM-O, T-LESS, ITODD, HomebrewedDB, YCB-Video, IC-BIN and TUD-L, but their shared format does not make them interchangeable occlusion experiments (BOP contributors, 2026b). Table 2 describes what each family can reveal and the principal confounds. Dataset-level difficulty should not be interpreted as a one-dimensional severity ordering.

[Table 2 about here]

LM-O emphasizes cluttered known-object scenes, while T-LESS exposes the combination of limited texture, similar shapes and symmetries. The dataset introduced with the object-coordinate work supplies the historical basis for occlusion-oriented tests, and T-LESS explicitly documents challenging industry-relevant objects (Brachmann et al., 2014; Hodaň et al., 2017). A method advantage on these datasets can therefore reflect correspondence quality, identity discrimination or symmetry handling in addition to the amount of visible area. Isolating an occlusion mechanism requires conditioning or controlled comparisons within the dataset, not simply assigning a dataset an “occluded” label.

ITODD is particularly relevant to mechatronic assembly and bin picking because its objects, sensors and evaluation requirements were selected for industrial recognition (Drost et al., 2017). Missing depth, geometric similarity and multiple instances interact with clutter. HomebrewedDB instead makes variability in lighting, appearance and scalability explicit (Kaskman et al., 2019). YCB-Video supplies everyday objects in video sequences and is central to the PoseCNN evaluation protocol (Xiang et al., 2018). Extracted test keyframes remain related to their acquisition scenes; frame counts should not be treated as counts of independent real-world arrangements.

Recent benchmark extensions widen the application context. BOP-H3 introduces hand interaction and reference-video onboarding, and the documented BOP-Industrial group adds

industrial data with single- and multi-view evaluation (Nguyen et al., 2025; BOP contributors, 2026a). However, not all BOP-H3 data provide depth, and industrial images may be monochrome rather than RGB. We use these settings to mark the boundary of the RGB-D synthesis, not to collapse them into a common modality. Multi-view input can resolve hidden structure, but a gain over single-view input combines an estimator change with an observation change.

The useful unit of evidence is therefore a specified dataset release, sensor, split, target list and evaluation script. Original dataset results and a BOP subset can use different frames or eligibility conditions. Training from real target images, target-object synthetic renders and non-target synthetic objects also creates different learning conditions. Keeping these fields explicit is more informative than collecting a large table of headline accuracies without their provenance.

What pose metrics preserve and discard

Point distances and symmetry handling

Let X be the sampled model, $\hat{T}$ the estimated pose and T the reference pose. The average distance of corresponding model points (ADD) is the mean of $\|\hat{T}\,x - T\,x\|$ over X. The nearest-neighbor variant, often called ADI or ADD-S, instead averages the distance from each transformed point to its nearest point on the other transformed model. The mixed notation ADD(-S) or ADD(S) often means that ADD is used for selected asymmetric objects and ADD-S for selected symmetric objects. The object-specific choice must be supplied; the abbreviation alone is insufficient (Hodaň, Matas & Obdržálek, 2016; Xiang et al., 2018).

Nearest-neighbor matching is not equivalent to minimizing a coherent rigid symmetry transformation. In ADD-S, each point can choose a different nearest neighbor. A symmetry-aware metric instead requires all points to share an allowed symmetry transformation. This distinction follows from the order of the minimization and averaging operations and can matter for similar or repeated geometry. A small nearest-neighbor error therefore cannot be interpreted as an equally small error in a unique orientation or in a task-relevant feature.

A threshold such as 0.1 times object diameter is a normalized tolerance. Its physical size changes with the object. An area under an accuracy-threshold curve also depends on the integration interval, units, normalization and object weighting. A YCB-Video ADD-S area under the curve (AUC) and a LM-O ADD(-S) threshold accuracy are different summaries even when both are reported as percentages. Neither can be converted into a BOP average recall by rescaling its headline number.

Visible consistency and maximum deviations

BOP localization combines visible surface discrepancy (VSD), maximum symmetry-aware surface distance (MSSD) and maximum symmetry-aware projection distance (MSPD). VSD compares rendered distance maps over the union of their visible masks. Its step cost is zero only where both poses are visible and the distance difference is below a tolerance. MSSD minimizes the maximum 3D model-point discrepancy over declared global symmetries. MSPD performs the

corresponding maximum comparison after image projection (Hodaň et al., 2020). Their distinct targets are summarized in Table 3.

[Table 3 about here]

A simplified definition of VSD, using the masks and distance maps produced by the specified renderer and visibility test, is

$$e_{\mathrm{VSD}} = \frac{1}{|\hat{V} \cup V|} \sum_{p \in \hat{V} \cup V} c(p) \quad (2a)$$

$$c(p) = 0 \; if \; p \in \hat{V} \cap V \; and \; |\widehat{D}(p) - D(p)| < \tau; \; 1 \; otherwise \quad (2b)$$

The tolerance tau controls surface misalignment, while the visibility test has its own tolerance. The surface and visibility tolerances should not be conflated. If the union of visibility masks is empty, the inspected toolkit assigns VSD an error of one (BOP contributors, 2026d). VSD assesses visible geometry and does not use color. Texture-based ambiguity and visible-shape consistency can consequently give different answers. Its treatment of hidden regions is useful for observation consistency, but does not certify the orientation of an unobserved task feature. These are interpretation limits of the definition, not evidence of an implementation defect.

$$e_{\mathrm{MSSD}} = \min_{g \in S} \max_{x \in X} \|\hat{T}x - Tgx\|_2 \quad (3)$$

$$e_{\mathrm{MSPD}} = \min_{g \in S} \max_{x \in X} \|\pi(\hat{T}x) - \pi(Tgx)\|_2 \quad (4)$$

The function pi is projection with the relevant camera calibration. MSSD expresses a maximum spatial discrepancy under the declared equivalence; MSPD expresses a maximum projected discrepancy. Camera range and focal length affect how a spatial error appears in pixels. Neither score is automatically a task tolerance. Maximum deviation can be informative for manipulating a rigid part, but grasp success also depends on contacts, collisions, calibration, material properties and control. The operational relevance must be established by a separate task evaluation.

Recall and precision are not alternative units of one score

In the historical BOP localization protocol, per-error average recall averages correctness over prescribed thresholds; VSD also averages over surface tolerances. The dataset score averages the three error-specific recall summaries, and the core score averages dataset scores (Hodaň et al., 2020). The complete threshold grid and any image-width normalization belong to the evaluation record. Component scores are useful because the same aggregate can conceal a tradeoff between visible geometry and full-model displacement.

The newer detection protocol uses average precision (AP), based on confidence-ranked predictions and matching, with MSSD and MSPD as pose errors (BOP contributors, 2026c). It does not merely rename localization recall. False positives and the absence of per-image target identities alter the question answered by the score. A localization AR and a detection AP should consequently appear in separate columns and cannot be used to calculate a common

improvement percentage. The 2024 challenge report documents the transition in tasks and information (Nguyen et al., 2025).

Two orthogonal extensions deserve separate reporting. Image-wise ambiguity annotations change what counts as a valid pose explanation (Meden et al., 2026). Visibility-balanced averaging changes the relative contribution of observations with different visibility (Pakdamansavoji et al., 2026). A system can improve under one extension and remain unchanged under the other. Distribution comparisons must identify whether the output is one pose, a hypothesis set or a normalized distribution. A confidence score on one pose does not encode all these objects (Haugaard, Hagelskjær & Iversen, 2023).

Methods compared through their dependence on visible evidence

Architectural names alone do not explain occlusion robustness. A more useful comparison asks which evidence enters the estimate, where correspondence is established, and which stage can fail when evidence disappears. Table 4 groups representative methods by these dependencies. The proposed failure modes are mechanistic interpretations of the documented pipelines, not a ranking established by experiments in this review.

[Table 4 about here]

Voting and fusion distribute the evidence

PVNet predicts directions toward image keypoints and uses voting to recover their locations before pose fitting (Peng et al., 2019). PVN3D transfers the voting idea to 3D keypoints inferred from RGB-D observations (He et al., 2020). These mechanisms explain why a keypoint need not itself be visible for an estimate to be proposed: visible pixels or points can predict its location. The inference still depends on the remaining support being informative. If the surviving patch is geometrically or visually ambiguous, many agreeing votes can support a wrong hypothesis.

DenseFusion combines appearance and point-cloud features at the pixel level and uses iterative pose refinement (Wang et al., 2019). FFB6D exchanges information between appearance and geometric feature streams throughout representation learning, followed by keypoint voting and fitting (He et al., 2021). These methods offer different ways of using partial evidence. Their designs do not imply that depth supplies the same effective information on every dataset. Missing or corrupted depth can remove geometric support exactly where RGB observations remain visible. A modality ablation and a visibility-stratified analysis answer complementary questions.

Direct regression also deserves protocol-aware interpretation. GDR-Net uses geometry-guided intermediate representations to support monocular pose regression (Wang et al., 2021). It should not be classified as independent of correspondence geometry simply because its final stage predicts pose directly. More generally, learned intermediate geometry, explicit voting and rendering-based refinement are not mutually exclusive categories. For this review, the important distinction is which inputs and assistance a reported variant actually uses.

Rendering and foundation features change the priors

CosyPose combines single-view hypotheses with multi-view association and scene refinement, while its single-view components also appear in benchmark variants (Labbé et al., 2020). MegaPose generalizes rendering-based coarse estimation and refinement to novel objects, with RGB and RGB-D variants (Labbé et al., 2023). Rendering can compare a partial observation with many candidate model views, but its success depends on initialization, rendering assumptions and the evidence available to the scoring function. More hypotheses or iterations also consume computation, so time and candidate budget belong to the comparison.

FoundationPose supports model-based and model-free estimation and tracking through a shared framework (Wen et al., 2024). SAM-6D combines proposal generation and instance selection with RGB-D point matching for novel-object pose estimation (Lin et al., 2024). FoundPose instead establishes RGB image-to-model correspondences using foundation features and can be combined with refinement (Örnek et al., 2024). These differences affect which missing evidence can be compensated by a model, a reference observation or geometric alignment. They do not justify treating every reported variant as an identical "zero-shot" system.

Pipeline stage boundaries are particularly important under occlusion. Let D denote a relevant candidate reaching the pose stage and C a correct final estimate, with C implying D. Within a visibility stratum v, conditional probability gives

$$P(C \mid v) = P(D \mid v)\, P(C \mid D, v) \quad (5)$$

Equation 5 is a conceptual decomposition, not a fitted model. It shows why results with oracle masks cannot establish the behavior of a full detector-pose pipeline: they change the candidate availability term. Conversely, an improved detector can increase total recall without improving conditional pose inference. Reporting both total and assisted pose-stage results can localize the source of a gain, provided the assistance and association rules are explicit.

Multiple hypotheses preserve unresolved structure

Manhardt and colleagues explicitly model multiple pose explanations, and SpyroPose estimates distributions over rigid transformations (Manhardt et al., 2019; Haugaard, Hagelskjær & Iversen, 2023). An ambiguous image may not support a single defensible pose, even after feature fusion improves the remaining evidence. Retaining alternatives can help a downstream system decide whether another view is necessary. However, retaining more candidates does not automatically provide calibrated uncertainty. Evaluation must distinguish whether a set covers plausible solutions, whether its mass is allocated meaningfully, and whether a single selected output is accurate.

The same distinction limits the interpretation of benchmark top-k hypotheses. A procedure allowed to return many candidates has a different opportunity to match the reference than a single-output system. Any comparison of coverage must therefore state the output budget and matching rule. BOP-Distrib supplies an evaluation framework for pose distributions (Meden et al., 2026); its annotations and metrics should be identified explicitly rather than mixed with the ordinary single-pose leaderboard.

A worked audit of published comparison conditions

We applied the four-layer record to eight documentary cases (Table 5). The cases were chosen to expose distinct interpretive failures, not sampled to estimate their frequency. For each case the table names the observable protocol difference, the conclusion it permits and the artifact needed for a stronger comparison. No accuracy value was recalculated. The purpose is to make the framework operational using evidence already present in published evaluation descriptions.

[Table 5 about here]

The first two cases illustrate why even the same dataset and metric name are insufficient. PVN3D reports both its standard YCB-Video comparison and a table using ground-truth instance segmentation (He et al., 2020). The latter isolates a pose-stage question and should not be silently substituted for end-to-end performance. FFB6D reports YCB-Video AUC and separate LM/LM-O diameter-threshold results (He et al., 2021). A statement that one percentage is larger than another across those summaries has no shared error tolerance or evaluated population.

The rendering-based cases expose the limits of broad method labels. MegaPose explicitly distinguishes PPF/SIFT hypotheses from detections supplied by a target-trained Mask R-CNN in its BOP experiments (Labbé et al., 2023). Thus, novel-object generalization of the pose networks does not, for every reported row, establish novel-object generalization of the detector. FoundationPose likewise specifies perturbed ground-truth boxes in its model-free YCB-Video comparison (Wen et al., 2024). This is a valid assisted evaluation, but it cannot be read as an unassisted detector result. The correct remedy is to identify the row and assistance, not to reject the study as a whole.

Other differences are structural. The single-view benchmark variants of CosyPose and the original multi-view system use different observations (Hodaň et al., 2020; Labbé et al., 2020). BOP localization and detection use different information and summary statistics (Nguyen et al., 2025). BOP-Distrib alters the ambiguity annotations, whereas WALDO balances visibility strata (Meden et al., 2026; Pakdamansavoji et al., 2026). These comparisons can establish progress under a named protocol. They cannot establish a protocol-independent ranking of occlusion robustness.

For a new pair of studies, the audit begins with task, split and target-list identity, then checks V, E, I and A (Figure 2). If only the metric differs and complete pose predictions remain available, re-evaluation can sometimes resolve the mismatch. If one system used extra views or privileged masks, recomputing its score cannot remove that information advantage. If a required detector condition or symmetry convention is unknown, the appropriate outcome is an unresolved contextual comparison. Missing information is not evidence that the conditions agree.

[Figure 2 about here]

Reporting recommendations and research implications

A compact record for defensible claims

Table 6 provides a minimum reporting record for authors and reviewers. Most fields already exist across BOP files, leaderboard metadata and method papers; the recommendation is to keep them together when making an occlusion claim. The record is intended to reduce ambiguity, not to define a new compliance score. A blank machine-readable version is supplied in Supplementary File S3, alongside the worked audit.

[Table 6 about here]

The official benchmark aggregate should remain available for continuity. We recommend adding performance by declared visibility intervals, the number of eligible instances in each interval, and error-specific components. Joint stratification by visibility, object and depth validity is useful when support is sufficient, because an apparent visibility trend can otherwise reflect a changing object mix. Sparse or empty strata must be shown rather than assigned an arbitrary success rate. Broad bins are preferable to a detailed curve whose severe-occlusion tail contains too little evidence.

Let R_{b} be recall in stratum b, n_{b} its instance count and N the total count across B supported strata. A generic balanced visibility summary can be written as R_{bal} = (1/B) sum over b of R_{b} for a predeclared set of B supported strata. WALDO already instantiates visibility-decile averaging (Pakdamansavoji et al., 2026); we do not propose this expression as a new metric. Its estimand differs from the empirical-mixture average R_{mix} = sum over b of (n_{b}/N) R_{b}. The difference is a choice of target population, not proof that one result is universally unbiased. A balanced score depends on bin boundaries, support and the handling of empty strata. When comparing datasets, these choices must be harmonized or the scores remain conditional on different populations.

Uncertainty should be reported at a unit justified by the acquisition process. Resampling individual adjacent frames can understate variability when they share a scene. Scene- or sequence-level resampling is a useful option when enough independent groups exist, but it does not repair a limited set of objects or environments. This is a recommendation for future evaluation; the present review computes no confidence intervals. Claims about an architecture causing better occlusion handling require matched data, assistance, refinement and computation, preferably with ablations. A cross-paper difference is not such an ablation.

From perceptual correctness to manipulation relevance

For robotics and embodied intelligence, the ambiguity that matters depends on the action. A rotation around a cylinder axis may be irrelevant to a broad grasp and decisive for inserting a keyed feature. A low VSD can be consistent with an unseen misaligned handle, while a large raw rotation difference can be harmless for a symmetric part. These examples follow from the different targets of pose metrics; they are conceptual cases rather than observed task outcomes. Pose evaluation should therefore be paired with explicit action tolerances or downstream trials before claims extend to manipulation reliability.

A useful research direction is to connect benchmark equivalence with task-conditioned equivalence while keeping both visible. The perception benchmark would retain its standard score, while a separate task evaluation would state which pose alternatives preserve feasible contacts or insertion geometry. Distribution estimators could then be assessed for both observation coverage and action-relevant uncertainty. BOP-Distrib and SpyroPose provide relevant foundations, but neither eliminates the need to specify the action and validate the decision rule (Haugaard, Hagelskjær & Iversen, 2023; Meden et al., 2026).

Another priority is testing what information is missing, not only how much. Removing a symmetry-breaking region and removing an equally large redundant patch need not induce the same uncertainty. Controlled occluder placement, matched visible fractions and explicit texture/geometry conditions could distinguish these mechanisms. Existing RGB-D data can support exploratory stratification, but a causal account would require a design that controls the confounds. Such studies remain future work; they are not claimed as results of this review.

Limits of the synthesis

The framework has not been evaluated for inter-reviewer reliability or tested as a predictor of ranking changes. The audit covers representative cases and cannot estimate how often the wider literature makes invalid comparisons. The targeted search may miss relevant work, especially outside English-language and openly accessible sources. Reliance on BOP-associated settings favors established rigid-object benchmarks and does not establish transfer to deformable objects, dynamic manipulation or continuous tracking. Current documentation was frozen at the consultation date; later dataset and toolkit changes require a new record.

Our conclusions are correspondingly bounded. The review supports explicit conditions under which an interpretation is defensible and identifies cases where a stronger inference is unsupported. It does not determine the best estimator, quantify the size of any protocol bias, or show that the proposed record improves scientific practice. Those questions need independent extraction studies, saved-prediction re-evaluation or prospective user testing. Making those missing tests explicit is part of the contribution.

Conclusions

Occlusion robustness is a conditional claim about an estimator, an observation and a measurement protocol. BOP provides a strong common foundation, but a dataset label and a headline score do not identify all conditions needed for comparison. Projected visibility, depth validity and truncation describe different observation properties. Global symmetry and image-dependent ambiguity define different equivalence targets. Assisted localization, full detection and altered score weighting also require distinct interpretations.

The four-layer record and worked audit provide a practical way to preserve these distinctions in literature synthesis. A defensible comparison either documents matching conditions, specifies the re-evaluation needed to establish them, or states the narrower contextual conclusion that the evidence supports. Future benchmarks should retain official scores while exposing supported

visibility strata, ambiguity assumptions and pipeline assistance. For robotic use, these perceptual measures should inform, and then be validated against, the requirements of the intended action.

Acknowledgments

Data and code availability

No new experimental data were generated or analyzed. The source evidence register, search log, documentary audit and blank comparison record accompany this manuscript as Supplementary Files S1-S3. Public source URLs and the inspected BOP toolkit revision are included in these files. The review does not redistribute third-party datasets or claim to reproduce benchmark results.

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Figure legends

Figure 1. Four layers connecting a partial observation to a comparable result.

Visibility describes what is observed; equivalence specifies acceptable pose explanations; information describes what the evaluated pipeline receives; aggregation determines the population represented by the score. A mismatch can change the interpretation even when the dataset name and score label agree. This is an original conceptual diagram, not a measured causal model.

Figure 2. Decision procedure for a proposed cross-paper comparison.

Check the evaluated task and population, then the four-layer record. A comparison can be direct, require a stated re-evaluation, or remain contextual. Recomputing scores can address some metric or weighting differences, but cannot remove extra information already used by a system. Missing fields remain unresolved. The procedure is proposed here and has not undergone an inter-reviewer reliability study.

Table legends

Table 1. Positioning against retrieved broad reviews.

Scope descriptions refer to the retrieved articles, not a claim that any topic is absent from them. The contribution is the joint comparison procedure. BOP-Distrib and WALDO are evaluation advances, not competing broad surveys.

Table 2. Benchmark families and the interpretation of occlusion evidence.

This is a qualitative map, not an ordering of dataset difficulty. Versions, sensors, splits and target lists remain necessary even when the dataset name is identical.

Table 3. Error functions and score summaries answer different questions.

Definitions: (Hodaň, Matas & Obdržálek, 2016; Xiang et al., 2018; Hodaň et al., 2020; BOP contributors, 2026c; Meden et al., 2026; Pakdamansavoji et al., 2026). ADD(-S) is used here as a generic mixed-criterion label, not as a distinct distance function. AP = average precision; AR = average recall; AUC = area under the accuracy-threshold curve.

Table 4. Methods grouped by their dependence on partial observations.

These are overlapping mechanisms, not mutually exclusive architectural classes. Dependencies and cautions are this review's interpretation of the documented pipelines. No ranking or causal effect is inferred.

Table 5. Worked documentary audit of eight comparison cases.

This is a purposive audit of protocol statements, not a re-analysis of numerical scores or a prevalence study. The remedy identifies what would be needed; its execution is not claimed. AR = average recall; AP = average precision.

Table 6. Minimum record for an occlusion robustness claim.

This proposed checklist consolidates requirements already dispersed across benchmark protocols and papers. It is not a validated reporting guideline. V, E, I and A correspond to the four layers in Figure 1.